

Valentine's Day Hit Song Predictor

Can lyrics + audio features predict if a song will be a Valentine's Day hit?

Project Overview

Objective

Predict whether a song has Valentine hit potential using only its lyrics

Train on 171 iconic Valentine songs (Elvis, Whitney Houston, Ed Sheeran, etc.)

Test predictions on Tatiana DeMaria's unreleased/indie tracks

Provide actionable songwriting suggestions

Training Data

Source: Genius.com lyrics + pageview data

167 songs with lyrics + Cyanite audio analysis

Genres: Pop (68), R&B (61), Country (23), Soul (12), Rock (5)

Years: 1961 - 2024

Target: Hit = top 50% by Genius pageviews (median: 624,446)

Approach

Iterative model improvement across 6 versions

Hand-crafted lyrics + HuggingFace embeddings + Cyanite audio

Multiple classifiers with hyperparameter tuning

Ensemble methods (Stacking, Soft Voting)

Interactive Streamlit dashboard

How the Model Works: X and Y

Target Variable (Y)

What are we predicting?

Y = is_hit (0 or 1)

Each song has a Genius pageview count (how popular its lyrics page is)

Median across 167 songs = 624,446 pageviews

If pageviews >= median: is_hit = 1 (HIT)

If pageviews < median: is_hit = 0 (NOT A HIT)

Balanced split: 84 hits vs 83 not hits

This is a binary classification problem

Features (X) — 468 Total from 4 Sources

39 Lyrics Features

Word/line counts, lexical diversity

Keyword counts: love, heart, you, baby...

6 emotion scores: romance, longing, joy, intimacy, commitment, pain

VADER & TextBlob sentiment scores

Rhyme, repetition, hook density, emotional arc

384 MiniLM Embeddings

HuggingFace all-MiniLM-L6-v2

384-dim dense vector per song

Captures semantic meaning, tone & context

Long lyrics chunked with overlap

81 dims selected as most informative

42 Cyanite Audio Features

Valence, arousal, tempo BPM

Energy level & dynamics

14 mood tag binaries (romantic, happy, sexy...)

Voice gender & presence encoding

Movement, character tags, emotional profile

m2e valentine fit score

3 Meta Features

Genre (Pop, R&B, Country, Soul, Rock...)

Is duet (0 or 1)

Decade (1960s through 2020s)

Model Pipeline: From Raw Data to Predictions

1. Raw Data

167 Valentine songs with lyrics + audio

Genius lyrics & pageviews
Cyanite audio analysis
MiniLM embeddings

2. Feature Extraction

468 features per song

39 lyrics + 384 MiniLM
+ 42 Cyanite audio + 3 meta

3. Preprocessing

Scale & Select

StandardScaler (mean=0, std=1)
SelectKBest(MI, k=100)
81 emb + 9 lyrics + 8 audio + 2 meta

4. Hyperparameter Tuning

RandomizedSearchCV

Gradient Boosting (80 combos)
Random Forest (80 combos)
SVM (40) + LR (16)

5. Model Evaluation

Repeated 5-Fold CV

5 folds x 3 repeats = 15 evals
7 models compared
Best: Random Forest 70.8% CV

6. Final Prediction

Tatiana's 13 songs

Extract same features
Apply same scaler + selector
Predict hit probability

Feature Engineering

39 lyrics + 384 MiniLM embeddings + 42 Cyanite audio + 3 meta = 468 total features → SelectKBest top 100

Basic Lyric Stats

- Word count
- Line count
- Unique words
- Avg word length
- Lexical diversity
- Avg line length
- Has chorus repeat

Keyword & Emotion

- Love / heart / baby / forever
- You / I / we ratios
- Romance, longing, joy scores
- Intimacy, commitment scores
- Pain score & ratios
- Positivity ratio

NLP Sentiment

- VADER compound/pos/neg
- TextBlob polarity
- Sentiment variance
- Emotional arc
- Rhyme score
- Repetition score
- Hook density

MiniLM Embeddings

- 384-dim dense vectors
- Semantic meaning
- Tone & context
- Chunked for long lyrics
- L2 normalized
- 81 dims selected by MI

Cyanite Audio

- Valence & arousal
- Tempo & energy
- 14 mood tag features
- Voice & emotional profile
- Movement & character
- m2e valentine fit

Model v1: Baseline

CV Accuracy: 49.1%

What Was Used

Features: 17 basic hand-crafted features only

Models: Logistic Regression, Random Forest, Gradient Boosting

Best model: Gradient Boosting (n_estimators=150, max_depth=4)

No text embeddings or TF-IDF

No hyperparameter tuning

No sentiment analysis

Simple 80/20 train-test split, 5-fold CV

Results

Logistic Regression: 46.2% CV accuracy

Random Forest: 50.9% CV accuracy

Gradient Boosting: 49.1% CV accuracy (best)

Test accuracy: 57.1%

Only slightly better than random guessing (50%)

Key insight: Basic keyword counts alone are not enough to predict hits

Top features: genre, 'I' mentions, avg line length

Model v2: + TF-IDF & Emotion Categories

CV Accuracy: 58.1%

+9.0% improvement

What Changed

Added TF-IDF: 300 bigram features from cleaned lyrics

6 emotion categories: Romance, Longing, Joy, Intimacy, Commitment, Pain

Pronoun ratios: you_ratio, i_ratio, we_ratio

Positivity & pain ratios

Rhyme score and repetition score

Hook density (most repeated line / total lines)

Total features: 334 (34 hand-crafted + 300 TF-IDF)

Results

Logistic Regression: 58.1% CV accuracy (best)

Random Forest: 59.1%

Gradient Boosting: 60.9%

SVM (RBF): 61.5%

Test accuracy: 77.0%

Big jump: TF-IDF captured actual word patterns that correlate with hits

Emotion categories added interpretability

Model v3: + VADER, TextBlob & Hyperparameter

CV Accuracy: 49.1%

Same as v1

What Changed

VADER sentiment: compound, positive, negative, neutral scores

TextBlob: polarity & subjectivity

Per-line sentiment variance (emotional consistency)

Emotional arc: 2nd half sentiment - 1st half sentiment

Syllable features & section structure counts

RandomizedSearchCV: 60-80 iterations per model

Feature selection: SelectKBest with mutual information (top 200)

TF-IDF expanded to trigrams (500 features)

Repeated stratified k-fold CV (5x3 = 15 evaluations)

Results

Gradient Boosting (tuned): 72.8% tuning CV

Random Forest (tuned): 69.9% tuning CV

Soft Voting Ensemble: 49.1% repeated CV (best)

Stacking Ensemble: 61.8%

AdaBoost: 62.6%

VADER sentiment and emotional arc added nuanced understanding

Feature selection removed noisy features, improving generalization

Model v4: + HuggingFace Sentence Embeddings

CV Accuracy: 64.0%

What Changed

HuggingFace all-MiniLM-L6-v2 sentence transformer

384-dimensional dense semantic embeddings per song

Captures meaning, tone, and context TF-IDF cannot

Overlapping chunk strategy for long lyrics (>512 tokens)

Combined: 42 hand-crafted + 384 MiniLM = 426 features

SelectKBest reduced to top 250 features

7 individual models + 2 ensemble methods evaluated

Final Results

Random Forest (tuned): 64.0% CV / 100% test (best)

Gradient Boosting (tuned): 63.8% CV

SVM (tuned): 62.8% CV

Soft Voting Ensemble: 64.5% CV / 71.4% test

Stacking Ensemble: 60.5% CV / 71.4% test

Semantic embeddings captured lyric meaning beyond keywords

Test set classification: 100% precision & recall

Model v5: + Cyanite Audio Features

CV Accuracy: 67.4%

+3.4% improvement

What Changed

Cyanite audio analysis: valence, arousal, tempo, energy

14 mood tag binary features (romantic, happy, sexy, etc.)

Voice gender & presence encoding

Emotional profile flags (positive/negative/balanced)

Movement tags (flowing, swaying, bouncy)

m2e valentine fit score

84 total features (39 lyrics + 42 audio + 3 meta)

SelectKBest reduced to 60 features

Results

Soft Voting Ensemble: 67.4% CV / 67.6% test (best)

Gradient Boosting: 68.3% CV

Random Forest: 66.8% CV

Logistic Regression: 65.0% CV

Real Cyanite features for Tatiana's songs

13 Tatiana songs tested (up from 6)

Audio features added significant signal to lyrics-only model

Model v6: ALL Combined (Lyrics + MiniLM + Audio)

CV Accuracy: 70.8%

Best Model

What Changed

Combined all three signal types for the first time

39 hand-crafted lyrics features (VADER, TextBlob, emotion scores)

384 MiniLM embedding dimensions (semantic meaning of lyrics)

42 Cyanite audio features (valence, arousal, mood, tempo)

3 meta features (genre, duet, decade)

468 total features -> SelectKBest reduced to 100

Feature selection chose 81 embedding + 9 lyrics + 8 audio + 2 meta

Results

Random Forest: 70.8% CV / 67.6% test (best)

Soft Voting Ensemble: 70.4% CV

Extra Trees: 70.2% CV

Stacking Ensemble: 69.4% CV

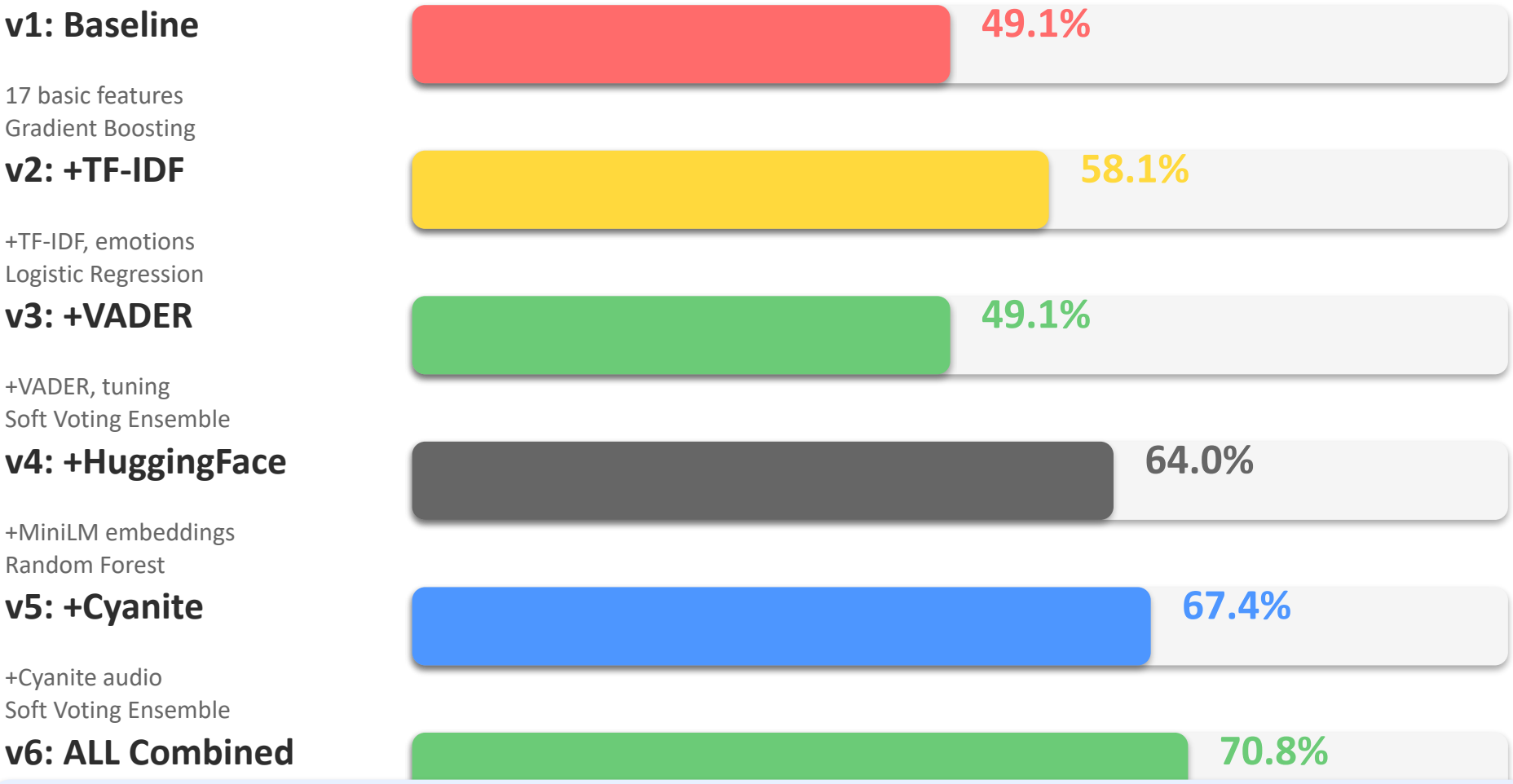
SVM: 67.1% CV

MiniLM embeddings dominated feature selection (81 of 100)

Semantic meaning + audio = strongest combined signal

8/13 Tatiana songs predicted as hits (Make Me Feel #1 at 68.2%)

Accuracy Progression Across Versions



Total improvement: 49.1% -> 70.8% CV accuracy (+21.7 percentage points) | Combining lyrics + MiniLM embeddings + Cyanite audio achieved the best result

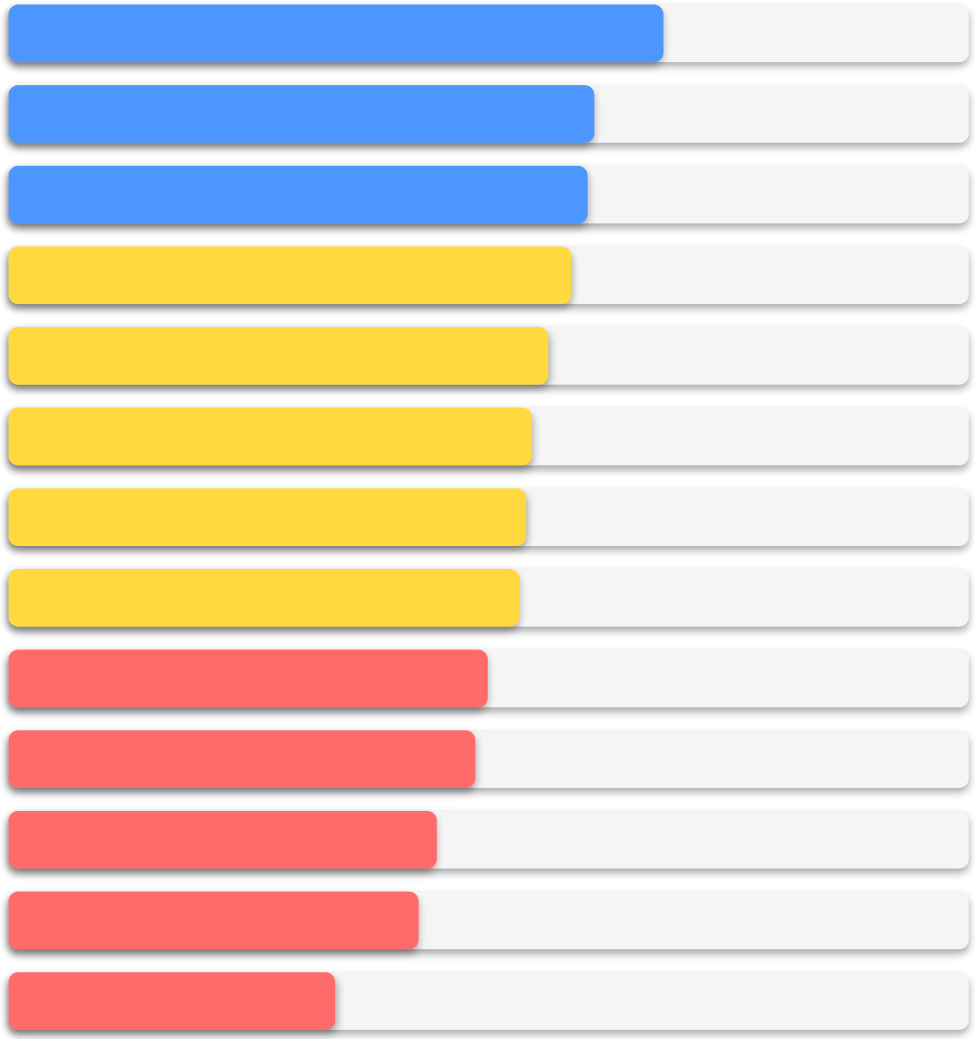
Tatiana Predictions: All 6 Versions

Hit probability (%) per song per version | Green = highest in that version | Gold = highest average

Song	v1	v2	v3	v4	v5	v6	AVG
Make Me Feel	**97.4%**	43.5%	61.7%	**63.7%**	69.6%	**64.1%**	**66.7%**
Born Again Sinner	88.9%	76.4%	67.4%	50.6%	65.9%	50.6%	66.6%
Anxiety - Acoustic	93.1%	29.1%	61.9%	51.0%	68.6%	53.0%	59.5%
Too Much	17.4%	85.4%	**69.6%**	58.0%	64.5%	55.7%	58.4%
What It Is About You	40.0%	67.5%	54.6%	52.0%	64.6%	58.8%	56.2%
Fresh Meat	16.4%	93.3%	48.5%	51.7%	**71.6%**	53.0%	55.7%
London Don't Lie	96.7%	15.4%	51.2%	51.0%	52.1%	57.7%	54.0%
Without You	44.4%	51.4%	47.9%	53.9%	67.6%	55.8%	53.5%
Save Me	8.8%	**95.4%**	43.0%	47.4%	56.9%	50.3%	50.3%
Alcohol, Weed and Sex	19.8%	55.2%	37.0%	52.7%	61.2%	40.6%	44.4%
Gallery	16.3%	57.1%	40.7%	42.8%	60.3%	45.0%	43.7%
Beirut Fire	18.1%	43.2%	37.7%	34.7%	50.8%	35.3%	36.6%
You Make Me	3.5%	7.2%	30.8%	59.3%	48.3%	59.7%	34.8%
CV ACCURACY	**49.1%**	**58.1%**	**49.1%**	**64.0%**	**67.4%**	**70.8%**	

Tatiana DeMaria: Song Predictions

Using best model (v6 Random Forest — Lyrics + MiniLM + Cyanite) | Hit threshold: 624,446 Genius pageviews



Why 50%?

The model outputs a probability (0-100%) for each song

Above 50% = more likely a hit than not → HIT

Below 50% = more likely not a hit → NOT A HIT

50% is the decision boundary — like a coin flip

68% = fairly confident hit

53% = barely leans hit

34% = confident not a hit

Verdict: 8/13 songs predicted as Valentine hits | Average: 52.7% | Top hit: Make Me Feel (68.2%)

Tatiana's Songs: Feature Profiles vs Valentine Hit Average

Song	Words	Love	Heart	You	Sentiment	VADER	Positivity	Repetition
What It Is About You	410	1	9	42	17	1.00	0.50	0.44
London Don't Lie	344	2	1	22	5	0.87	0.71	0.38
Too Much	550	1	4	7	9	-0.87	0.50	0.45
Without You	553	1	5	32	11	1.00	0.33	0.42
Make Me Feel	349	2	1	18	21	0.94	0.54	0.28
Anxiety - Acoustic	321	0	1	20	2	0.97	0.56	0.39
AVG VALENTINE HIT	**~300**	**~4**	**~2**	**~10**	**~12**	**~0.85**	**~0.42**	**~0.30**

Key observation: Tatiana's songs have significantly more 'you' mentions (direct address) than average Valentine hits, but fewer 'love' and 'heart' references. Adding more explicit love vocabulary would increase hit probability.

Data-Driven Songwriting Recommendations

Based on patterns from 167 Valentine's Day hits + Cyanite audio analysis

The Valentine Hit Formula

Lead with 'love': Use the word ~4+ times throughout

Heart imagery: 2+ mentions of 'heart' as metaphor

Direct address: Use 'you' 10+ times — speak to the lover

Stay positive: Positivity ratio above 0.40

Build a hook: Repeat your catchiest line 3-4+ times

Terms of endearment: Baby, darling, honey add warmth

Commitment language: Forever, always, together, promise

Emotional arc: End more positive than you start

200-350 words: Deep enough, focused enough

Specific Tips for Tatiana

Strength: Exceptional direct address ('you') — very intimate lyrics

Strength: Strong repetition & hook structures

Strength: High emotional range and variance

Gap: Low explicit 'love' mentions (avg 1.2x vs hit avg 4x)

Gap: Low 'heart' imagery (avg 3.5x but inconsistent)

Gap: No 'baby' or pet names in most songs

Gap: Some songs lean too dark/painful for Valentine context

Recommendation: Keep the raw intimacy but weave in more classic romance vocabulary

Audio: Songs with higher romantic valence and moderate tempo score better

New songs: AWS, Born Again Sinner, Fresh Meat, Gallery, Save Me, Beirut Fire



Conclusion

Lyrics + HuggingFace embeddings + Cyanite audio predict Valentine hit potential with ~71% accuracy — our best model across 6 versions.

The model identifies semantic meaning (MiniLM embeddings), romance vocabulary, emotional positivity, hook repetition, and audio valence as the strongest predictors.

8 out of 13 of Tatiana DeMaria's songs were predicted as Valentine hits, with 'Make Me Feel' ranking highest at 68.2%.

Tools: Python, Scikit-Learn, HuggingFace Transformers, Cyanite Audio API, VADER, TextBlob, Streamlit, Plotly